

Abstract

With novel datasets yet more precise observations the discussion of relation between earthquakes (EQs) and atmosphere in the form of seismo-tropospheric-ionosphere perturbations is becoming clear. The space based monitoring satellites has played a conclusive part to offer finding and study of anomalies ascribed to seismic developments. Rock deformation happens before occurrence of earthquakes due to stress within the breeding zone of EQs from the viewpoint of the physics this stress may cause electric currents in the rocks which may cause irregularities in atmosphere. The observing of atmospheric variances from modern instruments may provide some understanding about the anomalous nature of earthquake (EQ) before they occur. Tropospheric delays and Total Electron Content (TEC) estimated by GNSS receivers are studied in this paper within the seismic breeding zone over epicenter of Mw 7.1 (July 6, 2019), California. Similarly, continuous time series data of the Earth's lower atmospheric parameters like SO_2 and SO_4 are also investigated and they indicated abnormal variations before after the main shock of the EQ. Especially SO_2 which is remain disturbed for about 10 months even after the EQ. Furthermore, all these signs may authenticate the coupling between atmosphere and lithosphere over the epicenter within the breeding zone for the upcoming EQ.

Chapter # 1: Introduction

1.1 Navigation using Satellites

Global Navigation Satellite System (GNSS) is the arrangement of satellites in orbits especially Medium Earth Orbits. GNSS developed by the US (known as Global Position System-GPS), has 24 satellites revolving in orbits at an altitude of about 20,000 km. Several other countries have introduced schemes for the development of their own Regional or Global systems for navigation purposes. Beidou, GLONASS, Galileo are the important projects and changing the ways of positioning with modern navigation techniques and attaining more accurate, robust and reliable positioning like never before.

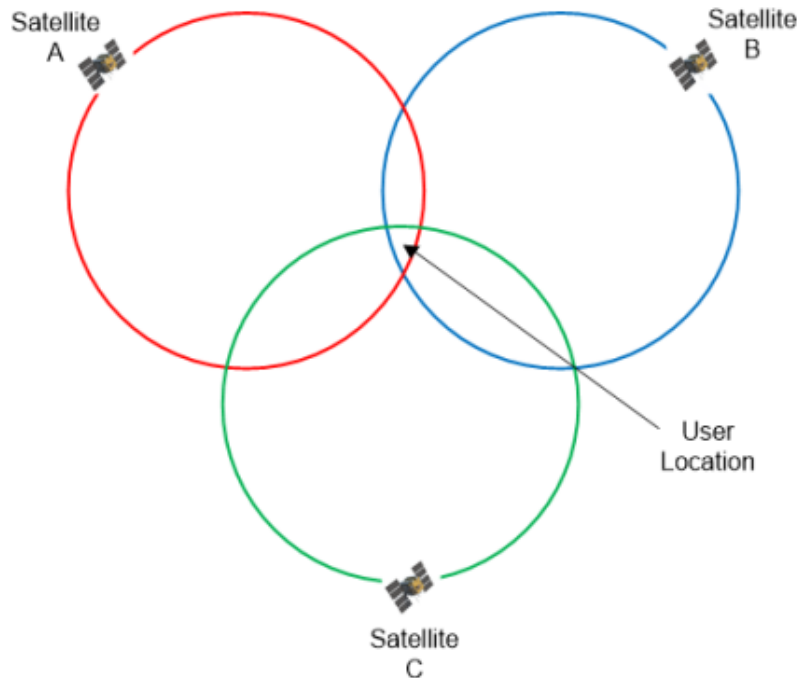


Figure 1: Trilateration required at least three satellites

The basic functioning of GNSS is that a continuous ranging codes is transmitted by the satellites at a certain carrier frequency, these ranging codes are then acquired and matched by

the GNSS receivers with a copy of known ranging code which identify the satellite is sending this code and estimated pseudorange of satellite from receiver is calculated.

1.2 GNSS Observations

Three measurements of incoming signals from satellites are necessary for the calculation of position, which are the following

1. Pseudo Random Code (PRN)
2. Carrier Phase
3. Doppler Measurements.

1.2.1 Pseudo Random Code (PRN)

By comparing or rather matching the received code with a copy of the code produced by the receiver, the pseudorange can be determined. It is the product of time required by GNSS signal to reach antenna and the speed with which electromagnetic waves travel. As a result of receiver clock and satellite clock synchronization errors, this distance is referred to as pseudorange.

1.2.2 Carrier Phase

The carrier phase is the number of waves between the transmitting antenna of satellite and receiving antennas. The number of complete wave cycles between the transmitting and receiving antenna are ambiguous so it only measures the variation in carrier phase.

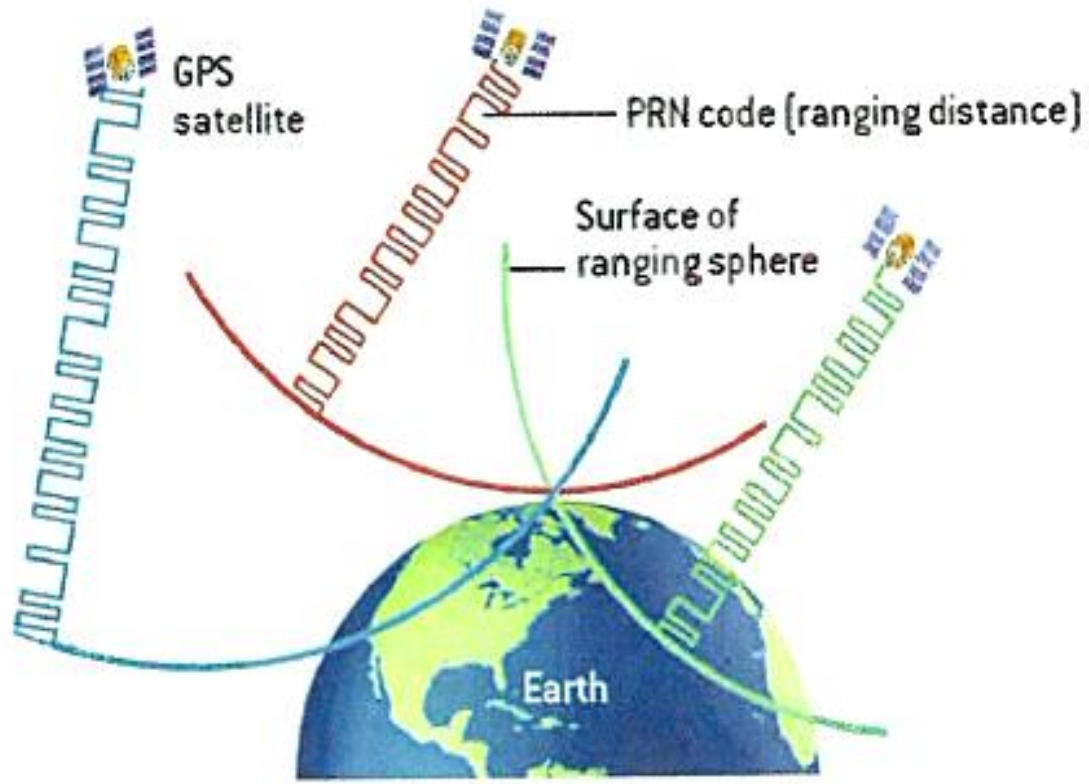


Figure 2: Pseudo Random code for distance estimate

Courtesy: <https://sourceforge.isae.fr/>

GPS satellites transmit signals on 1575.42 MHz and 1227.60 MHz other modern frequency are also added. Currently, the C/A code on the L1 frequency is public, while L2 frequency is encrypted which is only for military users.

Using dual-frequency measurements, we can better estimate the ionospheric error. Besides the ionospheric error, many other noises and errors which can affect the positioning solution. The measurement equation of pseudorange for GNSS is as follows:

$$p = \rho + d\rho + c(dt - dT) + d_{ion} + d_{trop} + \varepsilon_p \quad (1)$$

Where

ρ and p are the Geometric distance (m) and Measured Pseduorange (m) while dt and dT are the clock errors of satellite and receiver respectively.

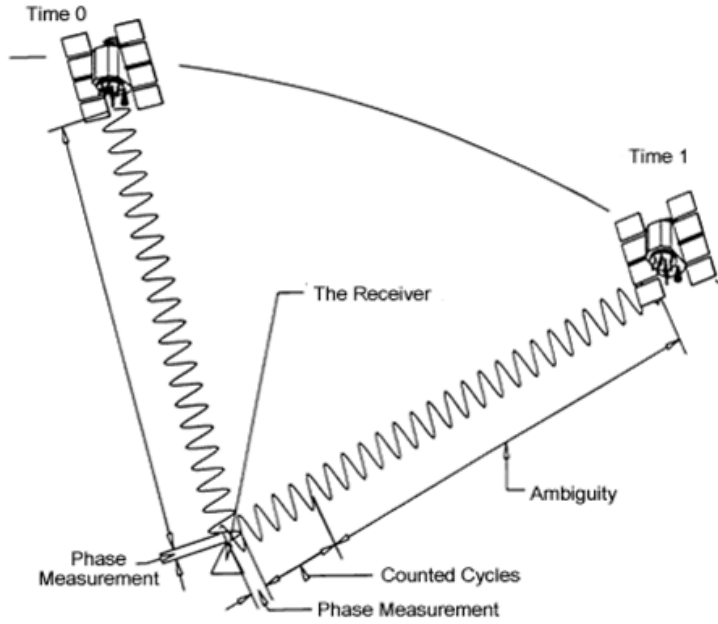


Figure 3: Phase Ambiguity of GNSS Signals [1]

The phase of the carrier can be measured with equation as follows.

$$\Phi = \rho + d\rho + c(dt - dT) + \lambda N - d_{ion} + d_{trop} + \epsilon_{\Phi} \quad (2)$$

where

Φ is the carrier phase (m), λ is carrier wavelength, N is ambiguous number of integral waves cycles), and ϵ_{Φ} is multipath error and noise. There are some similarities and some differences between this equation and equation 1. The integer ambiguity in phase equation is the most prominent dissimilarity. Additionally, there is a difference in magnitude of

ionospheric errors, which is the same for equations 1 and 2, however, they have the reverse sign. Because the ionosphere delays code measurements, the true range is smaller than estimated range.

1.2.3 Doppler Shift in GNSS Signals

The GNSS Doppler shift is an estimate of frequency change due to relative motion of satellite and receiver; it is usually calculated in cycles per second. The Doppler shift measurement is almost within + 5 KHz to 5 KHz and have no integral phase ambiguity.

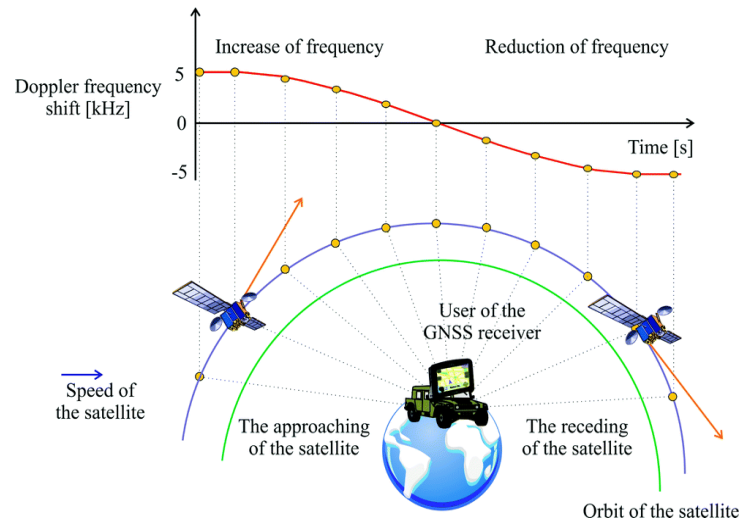


Figure 4: Approaching satellite signal's Doppler increases [2]

The equation of the Doppler is:

$$\dot{\phi} = \dot{\rho} + d\rho + c(\dot{dt} - \dot{dT}) - \dot{d}_{ion} + \dot{d}_{trop} + \dot{\epsilon}_{\phi} \quad (SEQ Equation \setminus * ARABIC 3)$$

Where the time derivative is represented by the Dot.

1.3 GNSS Errors

1.3.1 GNSS Signal Delay due to Ionosphere

The ionosphere is an area of the atmosphere's ionised gases that impacts electromagnetic transmissions. The ionospheric delay, which differs on the L1 and L2 frequencies, is a key influence of the ionosphere discussed in this work, the amount of the error by the ionosphere can be calculated. The magnitude of a typical ionospheric inaccuracy varies with angle of satellite elevation and position of Sun with respect to Earth [4]. Dual frequency can be used to determine ionospheric delay [5] as follows:

$$d_{ion} = \frac{f_2^2}{f_1^2 - f_2^2} (\phi_1 \lambda_1 - \phi_2 \lambda_2 - (N_1 \lambda_1 - N_2 \lambda_2)) \quad (5)$$

where:

f_1, ϕ_1 and λ_1 is frequency, carrier phase and wavelength of L1 and f_2, ϕ_2 and λ_2 is frequency, carrier phase and wavelength of L2 while N is number of ambiguous waves (integral ambiguity).

Relation mentioned above relies on the carrier phase ambiguity. To obtain the absolute ionospheric error, this should be computed. However, the ambiguity might be overlooked in order to determine the variations in error across time. These ionospheric inaccuracy is directly related to the total electron content (TEC), which varies throughout the solar cycle. [6].

1.3.2 Tropospheric Error

The part of the atmosphere which extends from surface to sixty kilometers above the earth's surface is the troposphere. Almost 90 percent of the delay in this part of signal path is due to the wet troposphere which is very difficult to model due to the variations in the tropospheric conditions. Factors like humidity, temperature, pressure, and elevation of satellite are main parameters to effect the wet delay [7]. The other component is the dry delay mainly caused mostly by gases such as nitrogen (N) and oxygen (O). At the zenith, the delay

due to dry portion is about 2.30 meters, while the delay due to wet can have value between 1.00 cm to 80 cm [1] and these values are almost ten times greater when the satellite is at horizon[7].

The delay due to troposphere is significantly reduced in differential positioning with short distances between two receivers. But, with a long distance between receivers the tropospheric error increases which leads to the position errors. Due to the non-dispersive nature of the troposphere at GPS frequencies, an approximation of the tropospheric impact requires the use of a model. Some useful models are: the Saastamoinen model [7], the Hopfield model [7], and the modified-Hopfield model [8].

Janes et al. [9], founds that the modified model brings the finest results for low elevation satellites than Hopfield model. For satellites with elevations greater than 20 degrees, all of these models provide the same results. The dry component of the tropospheric delay may be accurately approximated using a model to a few millimetres, while the wet portion is much less reliable.

A mapping function which is most accurate and easy to implement had been developed by Arthur Neill in 1996 [10]. Arthur showed there are the dependencies on surface temperature which limits performance of all pervious mapping functions

Neill mapping function is not reliant on meteorological parameters. He validated this claim by comparing this mapping with the data calculated with the help of radiosonde for duration of one year at various altitudes above sea level and covering wide area In order to validate this mapping function

Chapter # 2: Literature Review

The amount of energy released by the Earthquakes in the form of mechanical waves makes them one of the endangered phenomena that occur naturally. It always had been a wish of researchers to predict the Earthquake (EQ) even before realizing this destructive power to loss of life and mitigating infrastructural damage. Earthquake-related precursors have been described in many past studies but with advances, satellite constellations equipped with sensitive sensors have increased the accuracy of time coverage and area coverage. Now Scientist has more data to analyze atmospheric disturbances right before and after EQs and understand physics for possible modeling of lithospheric atmospheric modeling.

Programs like United States Geological Survey, UNAVCO, DEMETER, NASA, GOES and many others have been utilized in the study of Earthquake precursors [13]–[15].

Studies on the EQ-induced anomalies (electron density enhancement or depletion) in the ionosphere have validated the hypothesis of Lithosphere Ionosphere Atmospheric Coupling (LIAC) [16]–[23]. Tropospheric anomalies are also reported which are confirmation of the LAIC hypothesis [24]. Furthermore, [25] shows evident concerns in magnetic and electric fields before EQ [25]–[28]. Thermal anomalies further reinforced the LIAC hypothesis which was informed in studies related to Earthquake-induced anomalies in the atmosphere [15], [22], [23], [26], [29]–[35].

The tectonic motion of two slabs of Earth involves a large number of forces and parameters, depending on different microscopic and macroscopic phenomena the physics governing the tectonic motion becomes very complex. On the other hand, the vulnerable nature of the ionosphere and troposphere make them uncertain, and difficult to compare them with EQs for possible conformation LIAC hypothesis [36]. Even though the relation between EQs and

environmental anomalies (troposphere and ionosphere) is very ambiguous, many researchers have proposed some explanations during quiet geomagnetic conditions [37].

One concept among researchers is the emission of radioactive radon gas from deformed rocks due to pre-EQ stress [37], and another is the induced charges in stressed rock (positive holes) due to the deformation of rocks [24]. In the first concept, the emission of various gases especially radioactive radon which raises from the ground in the breeding zone of future EQ results in ion-molecular chain reactions which result in ion clusters or clouds generated in the atmosphere above the EQ breeding zone, followed by irregular variations in vertical electric fields, surface temperature, humidity nearby seismogenic zone. When the vertical electric field increases to a certain threshold where it is strong enough to affect ionospheric properties. This results in a variation of electron density in the ionosphere. The second approach was established with the concept of stress-induced holes generation in the EQ breeding zone, followed by ionization of the ionosphere and troposphere near the lithosphere [24].

Various experiments in the laboratory have shown the release of electromagnetic radiation from stress rocks but there is no solid mathematical validation mechanism to verify these concepts [38]. However a dense network of sensors and satellites constellation increases our insight into the hypothesis, the data from these satellites and sensors need to be checked for this hypothesis.

2.1 Research Objective

In this research, the GNSS tropospheric delay have been selected to detect any possible anomalies, tropospheric parameters like SO₂, Geopotential Height and OLR along with GNSS-TEC to understand possible explanation of the lithosphere-atmosphere coupling.

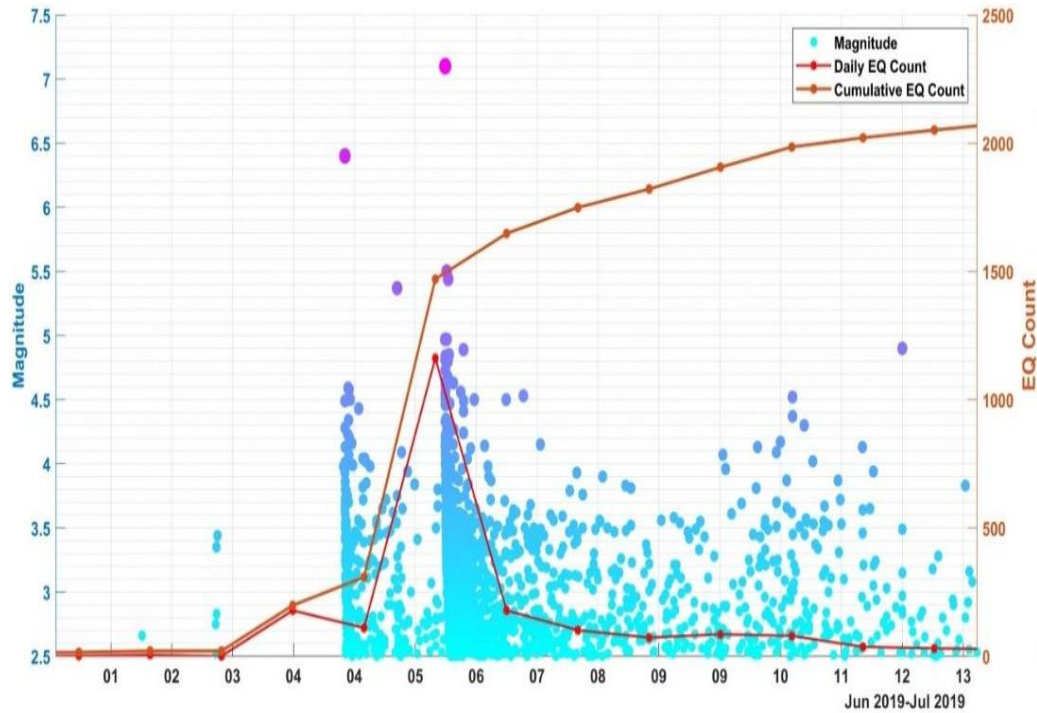


Figure 7: Sequence of Earthquakes and Daily Count

The study area in this present study in Searle's valley in California near Ridgecrest, this area was rattled by series of EQs on July 2019. The main event of magnitude 7.1 on July 6 was activated by more than 1000 small EQs and one major EQ of magnitude 6.4. The energy was huge that it creates new fault lines and surface the seismic waves were felt all over California. This event is ideal to study LAIC model due to occurrence of almost 30,000 foreshocks with Magnitude greater than 1, including more than ninety events with magnitude greater than 4 in the time span of almost 10 month after main event.

Statistical formula which describes seismically active area of potential EQ by [40]. If an Earthquake of Magnitude M occur than it may effect the cicular area of radius D as follows.

$$D = 10^{0.5Mw} \quad (7)$$

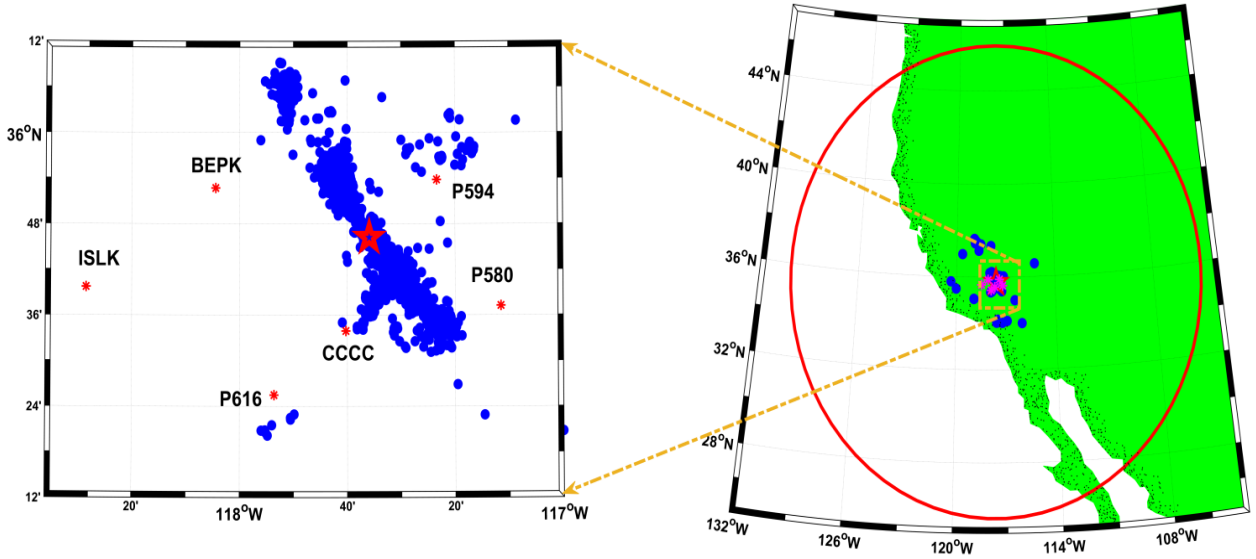


Figure 8: Selected Station in the breeding zone

It is desirable to choose GNSS station with in radius D as it is more probable that station outside this radius is unaffected by EQ stress-induced abnormalities. There are several of GNSS Stations installed by UNAVCO within the region.

2.2 GNSS Tropospheric Monitoring

The atmosphere of Earth is not uniform its mass varies inversely with respect to the altitude due to this inverse proportion of mass the electromagnetic wave deviates from the straight line as the refractive index keep changing with height. There are two components of atmospheric refraction N that is Dry and Wet component which are denoted by N_d and N_w respectively [7]

$$N = N_d + N_w \quad (8)$$

Unlike Ionosphere, troposphere is neutral electrically but the variation in the various gas concentration which is the function of height, affects the refractivity of the atmosphere. Due to the change of refractivity with respect to height the GNSS pseudorange estimate become bit off from the actual value its means one can estimate tropospheric parameters like water vapor by using tropospheric delays.

This is the reason many Environment Scientists are using now trying to use GNSS tropospheric error to monitor weather parameters.

The physics behind the method is not too complex as Earthquake. The electromagnetic wave passing through any medium in this case troposphere experiences some bending or rather delay in its travel time due to Snell's Law. With this the humidity in air and water can be detected which is in the form of gas in air with hourly time resolution. GNSS based estimates on weather parameter are already lining up with many meteorological models. But tropospheric estimates also have engineering challenge with it, the obligatory conditions are accurate phase ambiguity resolution along with the accurate time with very important for millimeter level accuracy of tropospheric delay as this is the level of accuracy that allow tropospheric estimates to monitor tropospheric variations. New GNSS frequencies are allowing end user to use multi frequency receiver to accurately estimate error like tropospheric delay and total electron content in ionosphere.

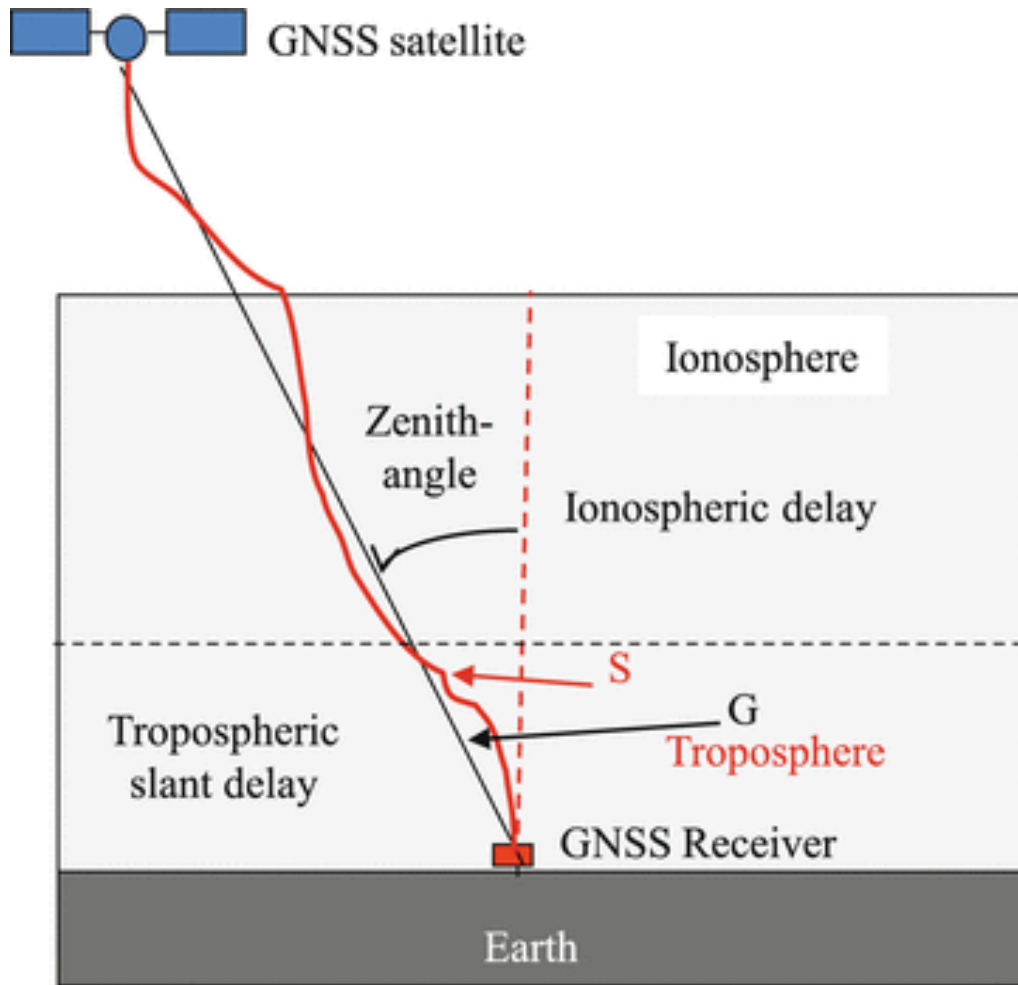


Figure 9: Tropospheric and Ionospheric Delays [41]

The tropospheric delay along the ray path is known as slanted tropospheric delay, which is then mapped to obtain Total Tropospheric delay in zenith direction.

2.3 GNSS Ionospheric Monitoring

The ionosphere only contains a tenth part of the atmospheric mass and its height is roughly from 60 km to 1000 km. As this part of the atmosphere is not electrically neutral, it has a significant contribution in atmospheric electricity regardless of its lesser contribution in atmospheric mass. Depending on the chemistry and concentration of ions as a function of height, the ionosphere is divided into four major layers, namely, D, E and F (includes F1 and F2).

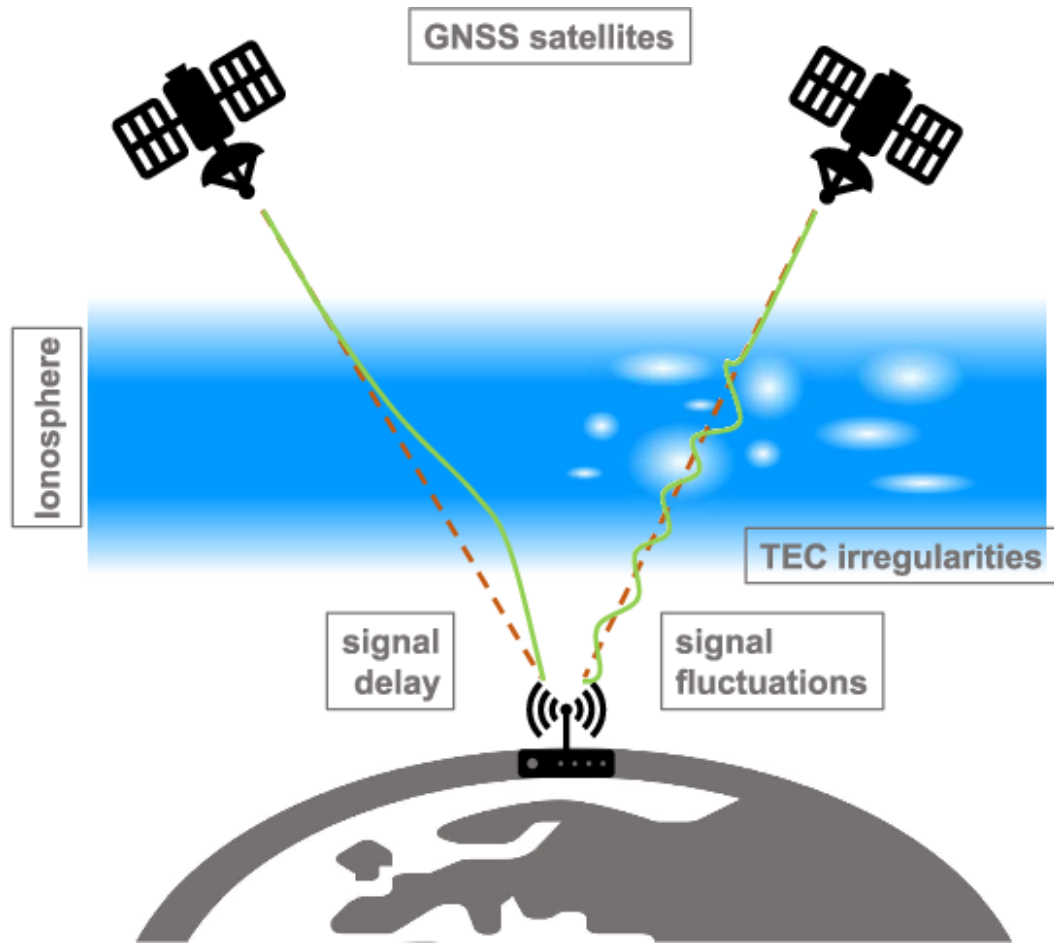


Figure 10: Ionospheric Delays [42]

2.4 Data Collection

UNAVCO provide GNSS Navigation and Observation files in study area of current research. These files are used to calculate Total Electron Content (TEC) with MATLAB program contributed by IONOLAB. Furthermore, NOAA provided Outgoing Long Range Radiations. The long term time series data for SO_2 and SO_4 is acquired by GIOVANNI, UNAVCO also provided tropospheric data which is processed by GAGE Analysis Center.

Chapter # 3: Methodology

3.1 GNSS Based TEC

Total Electron Content is measured in TECU which is equal to $10^{16} \text{electrons}/m^2$ [45] which is the number of electrons along the slanted ray track. If dual frequency receiver is used having signal frequency f_1 and f_2 then

$$STEC = \frac{f_1^2 f_2^2}{40.2(f_1^2 - f_2^2)} (P_1 - P_2 + \lambda_1(A_1 - A_2) - \lambda_2(A_1 - A_2) + \epsilon)$$

(SEQ Equation * ARABIC 9)

$$STEC = \frac{f_1^2 f_2^2}{40.2(f_1^2 - f_2^2)} (C_1 - C_2 - (d_1 - d_2) + \epsilon)$$

(SEQ Equation * ARABIC 10)

Where C and P are the code and carrier phase observation for the signal whose wavelength is λ and phase ambiguity estimated at receiver is A , the errors due to instrument is denoted as b . the pseudo distance between satellite and receiver is d , while random noise is denoted by ϵ

If the angle of satellite elevation Z is known than **STEC** can be mapped to Vertical TEC or **VTEC** with following equation where H is theoretical height of ionosphere and R is radius of Earth

$$VTEC = STEC \cdot \cos \cos \left(\arcsin \arcsin \left(\frac{R \cdot \sin(Z)}{R+H} \right) \right)$$

(SEQ Equation * ARABIC 11)

This method of calculating TEC is already implemented by [43], [44].

3.1.1 Abnormal Varations in Ionosphere

Solar activity is majorly responsible for the ionization or rather concentration of electrons in ionosphere. The solar activity can be estimated using Solar Flux and parameters like geomagnetic indices (kp & Dst). It is important that there must not any solar activity because

anomalies in ionosphere become ambiguous which will be discussed in next chapters in detail. TEC and VTEC anomalies are elaborated using confidence bounds depicted in following equation

$$\text{Confidence Bounds} = u(\text{TEC}) \pm 1.5 \sigma (\text{TEC}) \quad (\text{SEQ Equation } \backslash * \text{ ARABIC 12})$$

Deviations in TEC are defined as

$$\begin{aligned} DTEC = \{ & \text{TEC} - UB & \text{TEC} > UB & 0 & UB < \\ & \text{TEC} > LB - \text{abs}(\text{TEC} - LB) & \text{TEC} < LB & & \end{aligned} \quad (\text{SEQ Equation } \backslash * \text{ ARABIC 13})$$

The median of the data set is denoted by u , the Upper Bound UB and Lower Bound LB are determine by using standard deviation plus median of TEC in equation 12.

3.2 Tropospheric Delay Calculation

With the development of satellite technology many mapping functions have been adopted by engineers and researcher to mitigate effects of tropospheric refractivity. If slanted tropospheric delay (STD) is known then some mapping function $m(z)$ is used to estimate total tropospheric delay along zenith direction (ZTD).

$$ZTD = STD.m(z) \quad (\text{SEQ Equation } \backslash * \text{ ARABIC 14})$$

$$ZTD = ZHD.m_h(z) + ZWD.m_w(z) \quad (\text{SEQ Equation } \backslash * \text{ ARABIC 15})$$

where $m(z)$ is general mapping function, $m_h(z)$ and $m_w(z)$ are mapping function for the dry part and wet part for Zenith Hydrostatic and Wet (i.e ZHD and ZWD) Delay respectively

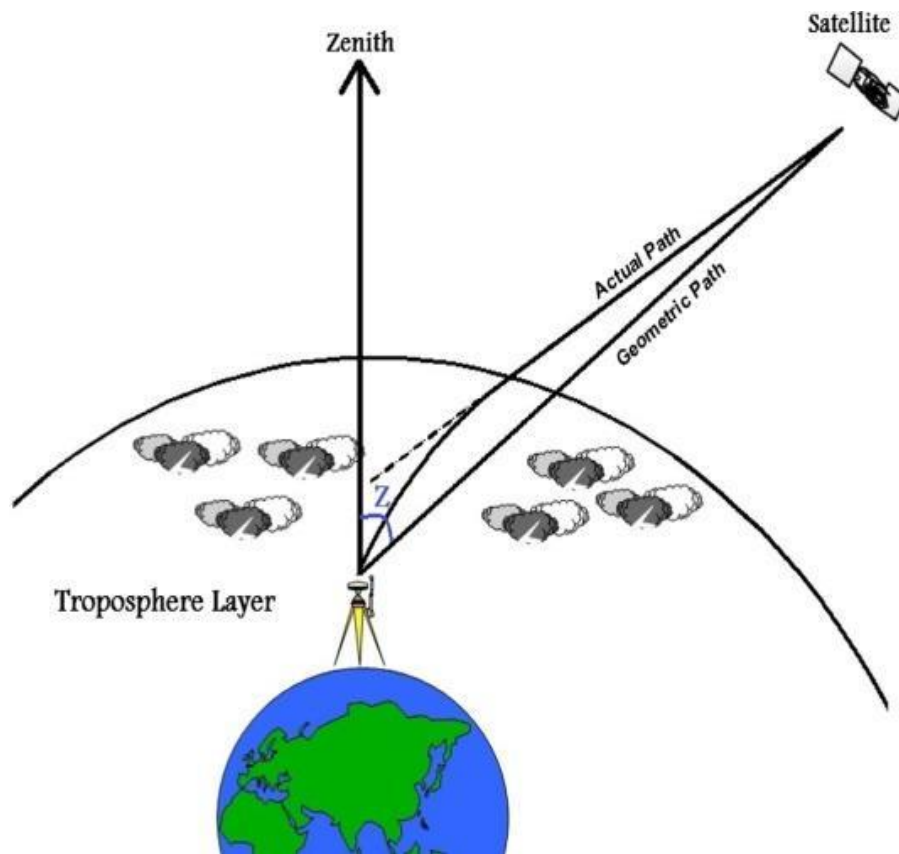


Figure 11: Tropospheric Zenith Delays
 Courtesy : <https://doi.org/10.1016/j.aej.2018.10.017>.

3.2.1 Neill Mapping Functions

A modern mapping function which can be used to estimate global wet and tropospheric delay is Neill Mapping Function which is function of elevation angle of satellite with respect to user receiver [10].

$$m_h(\varepsilon) = \frac{1 + \frac{a}{b}}{\sin \varepsilon + \frac{a}{\sin \varepsilon + c}} + \left[\frac{1}{\sin \varepsilon} - \left(\frac{1 + \frac{a_{ht}}{b_{ht}}}{\sin \varepsilon + \frac{a_{ht}}{\sin \varepsilon + c_{ht}}} \right) \right]$$

(SEQ Equation * ARABIC 16)

$$m_w(\varepsilon) = \frac{1 + \frac{a_{ht}}{b_{ht}}}{\sin \varepsilon + \frac{a_{ht}}{\sin \varepsilon + c_{ht}}}$$

(SEQ Equation * ARABIC 17)

Where

$$a(\varphi, t) = a_{avg}(\varphi) + a_{avg}(\varphi) \cos \cos \left(\frac{t-D}{365.35} 2\pi \right)$$

(SEQ Equation * ARABIC 18)

Here D is equal to 28 for North Hemisphere and 211 for South Hemisphere. The nearest guess for values of $a(\varphi, t)$ is linearly interpolated to start iterative process to calculate more accurate values for a in eq. Similarly b and c are obtained with same technique.

Least square fit to the height at nine elevation angles were used to determine correction coefficient for heights a_{ht} , b_{ht} and c_{ht}

As mentioned before tropospheric refractive index is larger than unity, it means the geometric path length will be less than radio math length for GNSS signals. The delay cause by the dry component is quite smaller on vertical excess length then vertical excess path caused by the wet component [46].

3.2.2 Tropospheric Anomalies

GNSS Tropospheric product by UNAVCO is calculated by algorithm which is based on Neill Mapping Function, the complete explanation of this algorithm is not in scope the current research. We analyzed the time series data for Tropospheric delays with time resolution of 5 min for UNAVCO station situated within the EQs breeding zone. The anomalies are detected using inter quartile ranges ($\mu \pm \text{IQR}$) over the two day running mean values of last two days. Tropospheric delay residue is then analyzed for different stations and almost similar anomalous behaviors have been observed before and after the main event. For the closer comparison among different stations two hours (2:00 UT to 4:00 UT) average of residue is plotted for each day for several sites and significant anomalous similarities is observed.

3.3 Lower Atmospheric Parameters

As mentioned before that vertical excess path for GNSS signal has little contribution from dry component of atmosphere but it does not mean that it is unaffected by the EQs. To investigate the dry part which mainly composed of dry gases like Oxygen, the area averaged is utilized from GIOVANNI database application in spatial grid on $(-122.8711, 26.4551, -106.875, 39.9023)$. The abnormalities due to Earthquake is analyzed using continuous wavelet transform

Chapter # 4: Result Analysis

EQ Forecasting means the location and time of future EQ is now to some level confidence due to this it is required to analyze the phenomena that are influenced by Earth's Crust and its tectonic stress furthermore, it is also required that these phenomena are time dependent. The changes in these phenomena can be linearly or non-linearly correlated. Research about Forecasting of Earthquake is usually focused on the mechanics of tectonic motion and it is an

agreement among researchers that on time scale level like meteorological forecasting, EQ forecasting is not possible in near future. In this study we have tried to correlated EQ with abnormal variation in troposphere

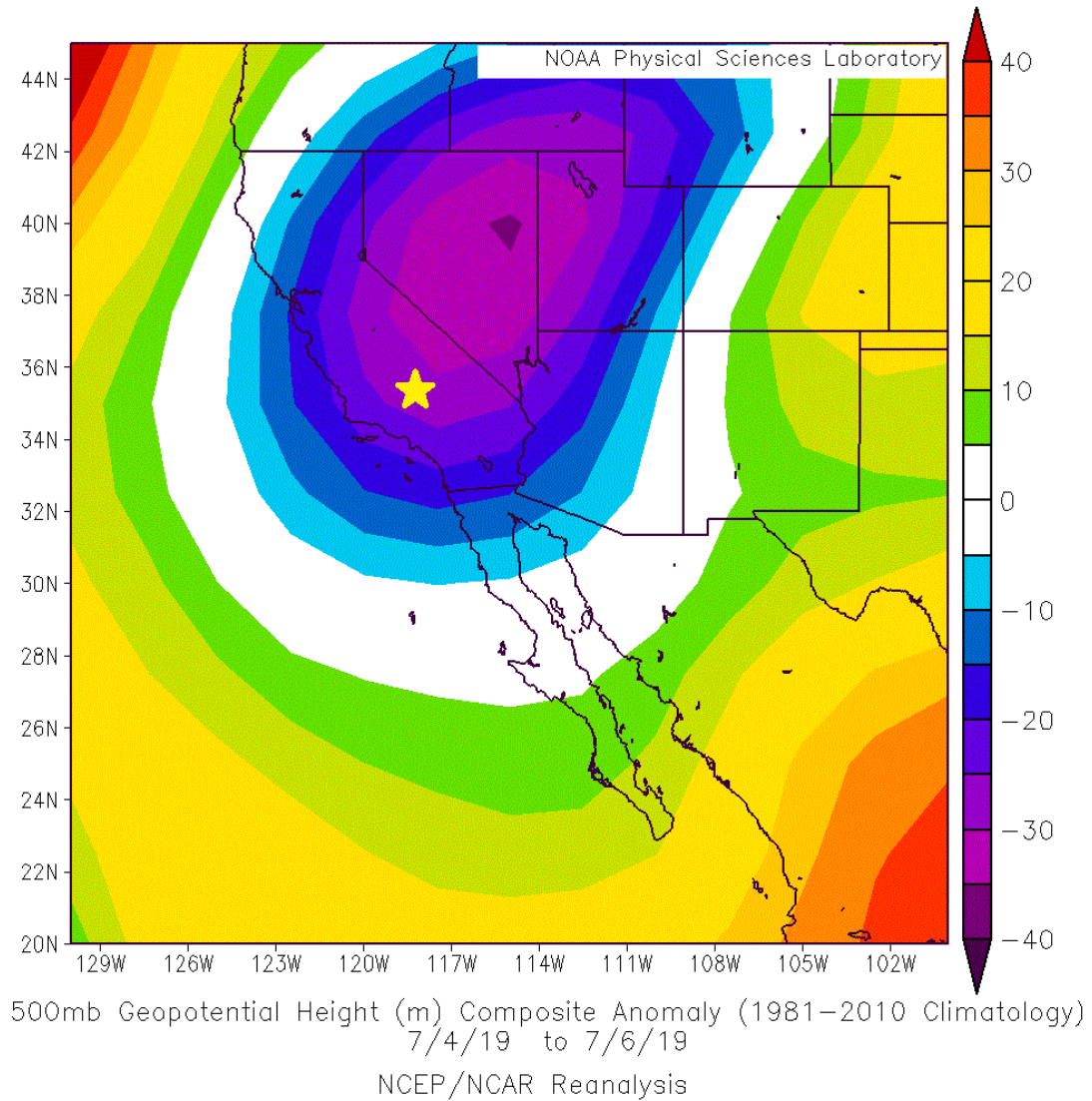


Figure 12: Geopotential Height (m) Composite Anomaly 7/4/19 to 7/6/19

4.1 Anomalous Geo-Potential-Height

Height of points having equal potential energy relative to sea level is known as Geopotential Height Z , the following expressions relates it with geometric height z

$$Z = \frac{1}{g_o} \int_0^Z g dz' \quad (19)$$

In this equation gravitation acceleration is represented as g and a global average of g at sea level is denoted as g_o . This parameter is reported in many studies regarding EQ, this important meteorological parameter shows significant correlation with seismic activities like EQ. For the present case, Geopotential Height decreases from the mean value during the time of EQ preparation and shows agreement with studies by Shah 2020 and Ouzounov 2011.

4.2 Anomalous Outgoing Long Range Radiations

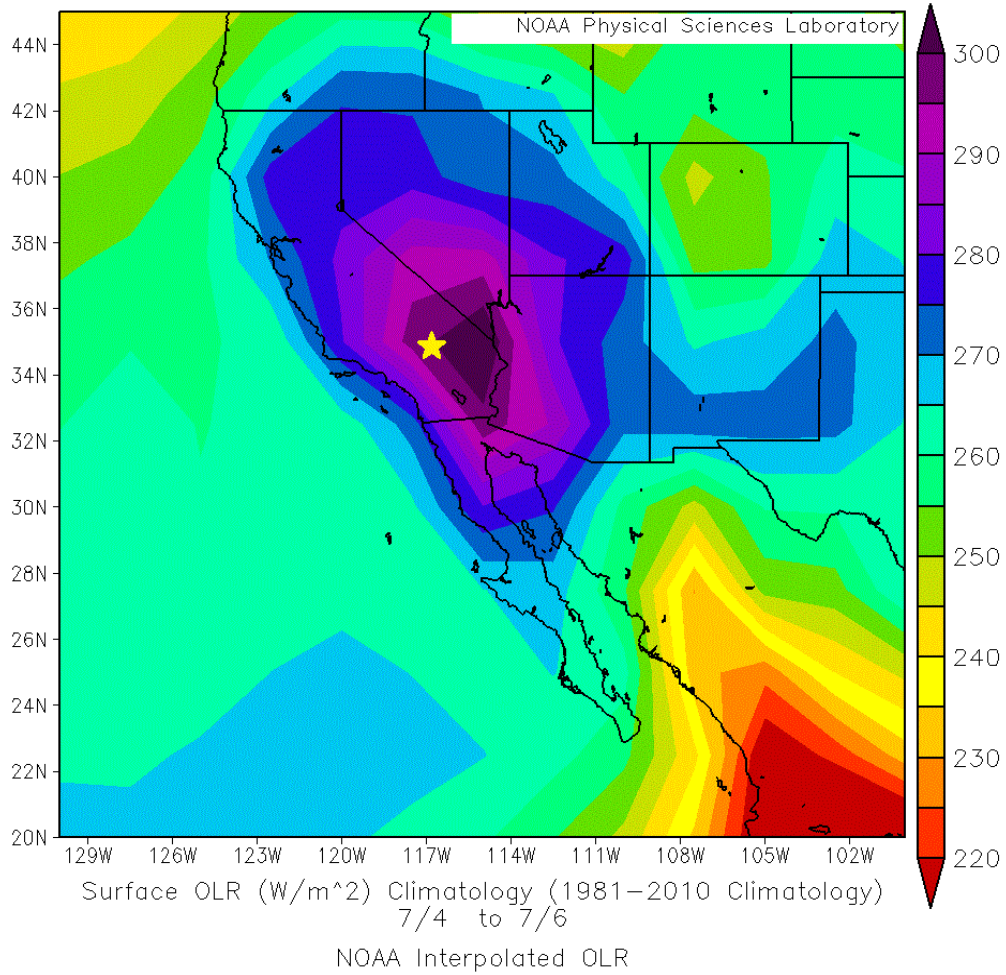


Figure 13: Surface OLR (W/m^2)

The thermal radiation which radiate from top of atmosphere and connects in general with Earth atmosphere are known as Outgoing Long Range radiations and could be triggered with contact between troposphere and Earth Surface.

These radiation shows abnormal variation during the period of seismic activity thus validating lithospheric and atmospheric interaction with agreement of past studies done by Ouzounov et al., (2011) shows that anomalous changes in OLR occur almost 5 days before the great Thouko Earthquake. Furthermore, a study cared by Shah et al. (2020) have claimed the same variation three days before three different EQs in Japan while other tropospheric parameter also show significant similarity. With this in mind we have also studies its variation during seismically active period during EQs and observed prominent anomalous variations.

4.3 Sulphur Dioxide Anomalies

The SO_2 concentration is related with abundant amount of oxygen ions in air. The analysis of time series data of sulfur dioxide by wavelet transform shows the obvious relation of SO_2 with seismic activities the abnormal behavior which starts few days before the main shock day and remain disturbed throughout the year.

This continuous anomaly in Sulpur Dioxide suggests that the presence of stress in the vicinity of breeding zone throughout the year after main shock. These variation are aligned and confirms with geospatial height and OLR. Furthermore, it also suggest the probaility of the oxygen ions.

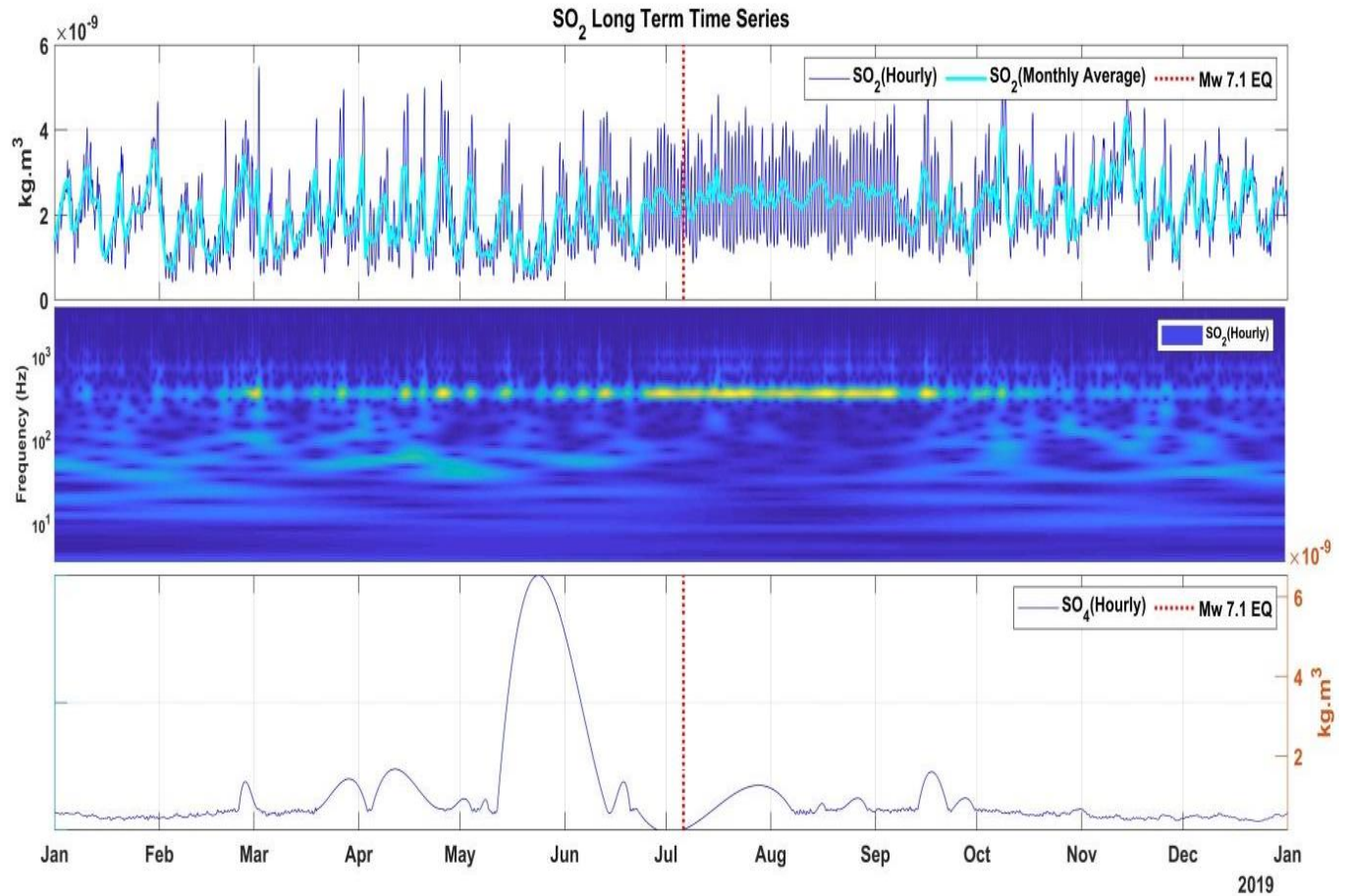


Figure 14: SO_2 time series (top), wavelet transform of SO_2 timeservers (middle row), SO_4 variations (lower row)

4.4 GNSS Zenith Tropospheric Delay

Analysis of OLR, Geopotential Height and SO_2 gave us some insight about relation between troposphere and lithosphere. But these parameter are directly or indirectly are the part of dry component of troposphere. To analyze wet component of troposphere GNSS wet tropospheric delay demand us to analyze it. Noticeable abnormalities in GNSS tropospheric delays observed within a day before the main event.

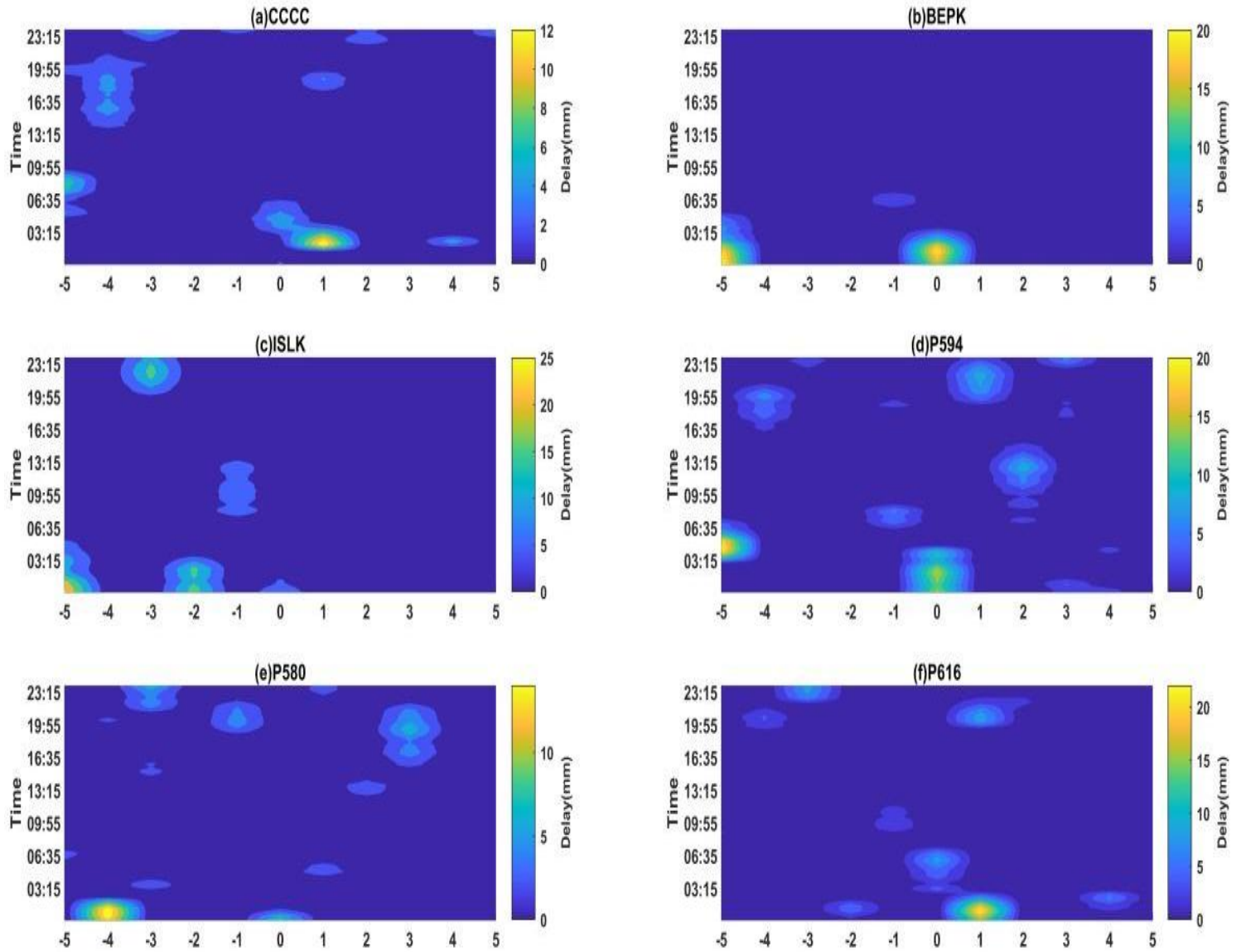


Figure15: Residue Zenith Wet Tropospheric Delay

Two hour average from 2:00 UT to 4:00UT (with in occurrence time of main shock) of residue of delays is plotted for different sites in the vicinity of EQ. the identical behavior of anomalies in tropospheric delays suggest that there is clear indication of interference of seismic activity in lithosphere with water vapours in Troposphere.

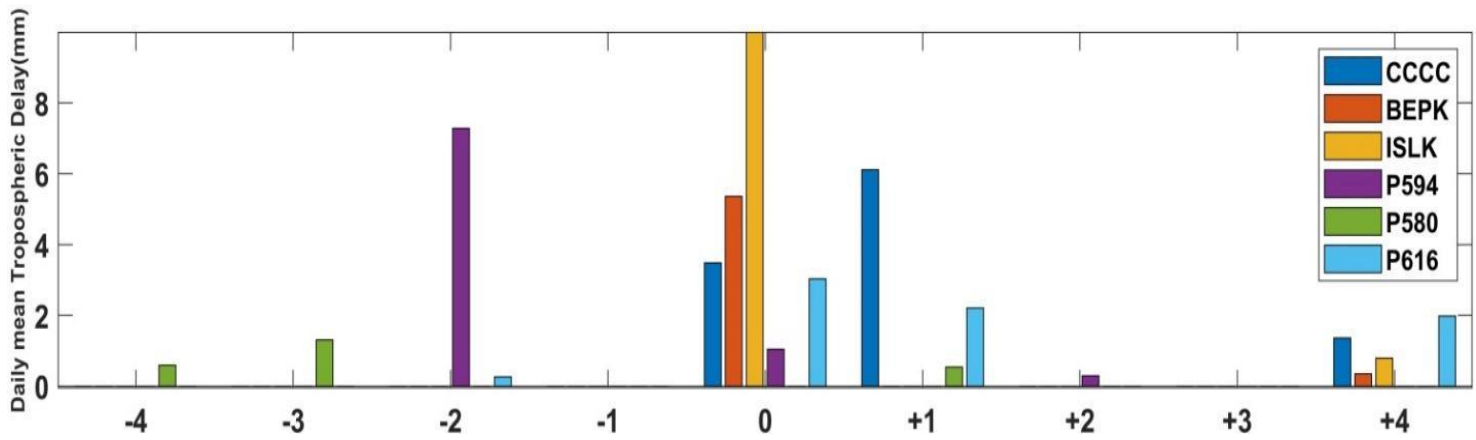


Figure16: Comparison of Residue Tropospheric Delay for different stations

4.5 TEC Anomalies

We have already mentioned about the previous studies which relates anomalies in ionosphere with EQs.

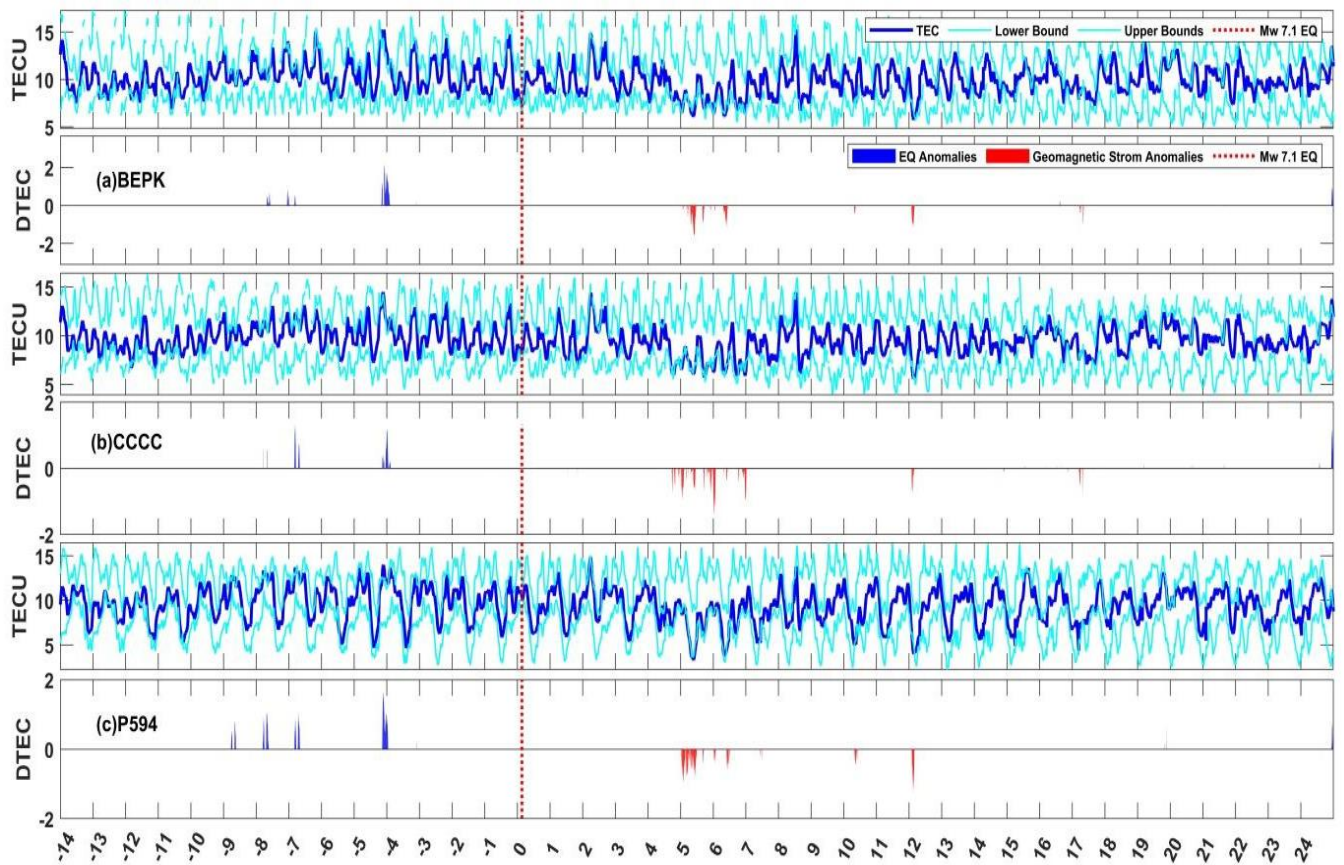


Figure17: Ionospheric Anomalies before EQ

In this study we also considered the GNSS based Total Electron Content and it is observed that ionospheric anomalies started almost 8 days before main shock and become prominently large until 3 to 4 days prior to main event this might be due to stress in the breeding zone of future EQ. The anomalies shown in red are not clear, it might be due to EQ or due to solar activities because the indices like Kp and Dst have larger values compared to the normal days this can be the indication of perturbed geomagnetic activities.

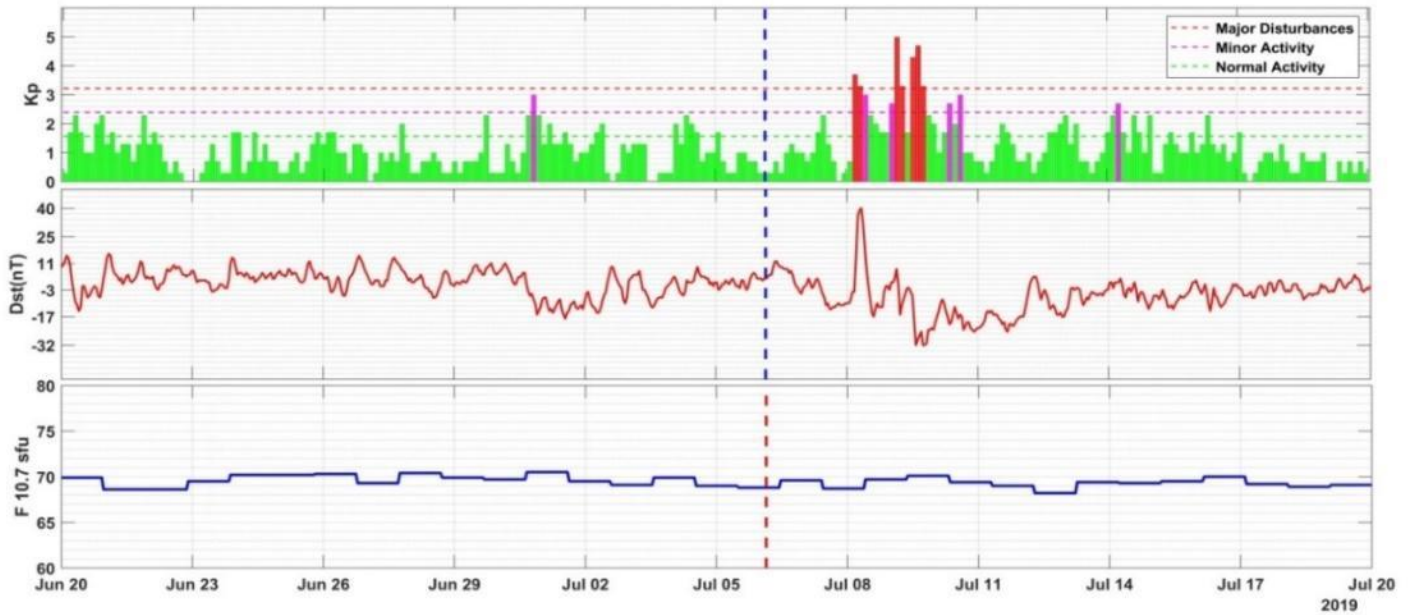


Figure18: Geomagnetic indices

Chapter # 5: Discussion and Conclusion

5.1 Discussion

Anomalies in ionosphere, GNSS tropospheric delay and other atmospheric parameters gave us the hint for possible correlation with EQs.

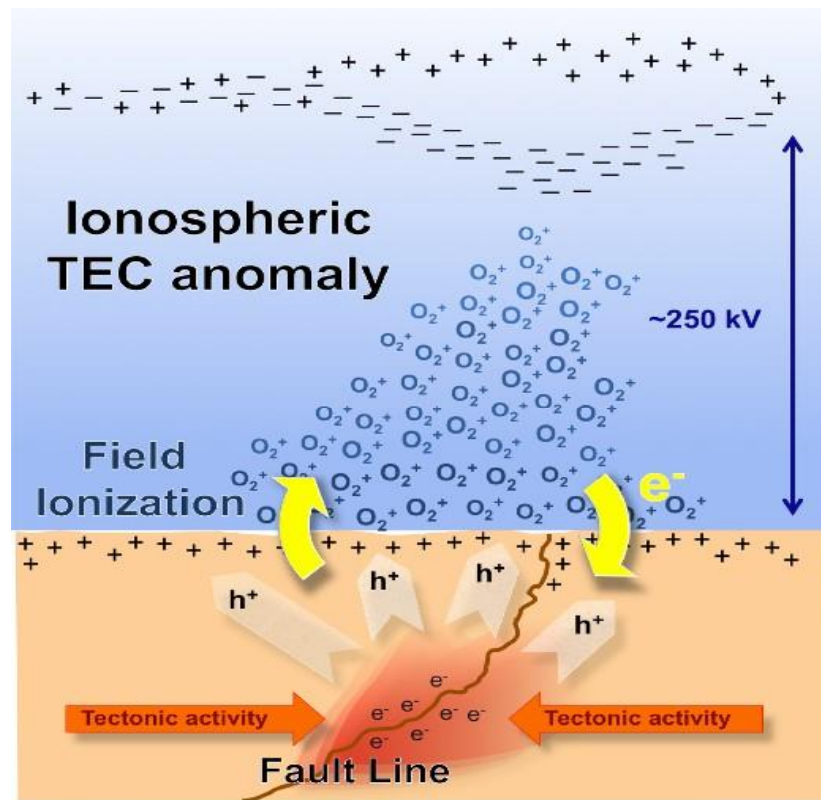


Figure 19: Physics of Tropospheric and Ionospheric variations due to EQ [47]

But there is no clear physical explanation for these anomalies until today. The interactions that are interfering ionosphere should be electrical in nature to reach that far, this idea suggest that everything starts with the ionization of air gases right above the breeding zone of future EQ which is due to induced stress on rocks. These naturally occurring rocks are not crystal prefect

in structure and have Peroxy defects, when some stress changes the deformation of these rocks results in generation of high mobility charges that positive holes (say p-holes). It is observed that these holes have ability to travel tens of kilometer but the ionization of air especially the gases with lowest ionization potential 12.07eV escalates when the surface potential cross the threshold value of +3V (Freund F. et al. 2009). The oxygen gas has lowest ionization potential among all abundant gases; its ions readily react with other gases as well.

Further development in stress also more positive holes to flow which result in surface potential variations, release of large ions at surface and spread of electrons on surface also recommend the possibility of corona discharge if flow of positive holes keep building up [24]. Which leads to a collapse of the positive potential and allow a lot of air electrons to rise to the surface

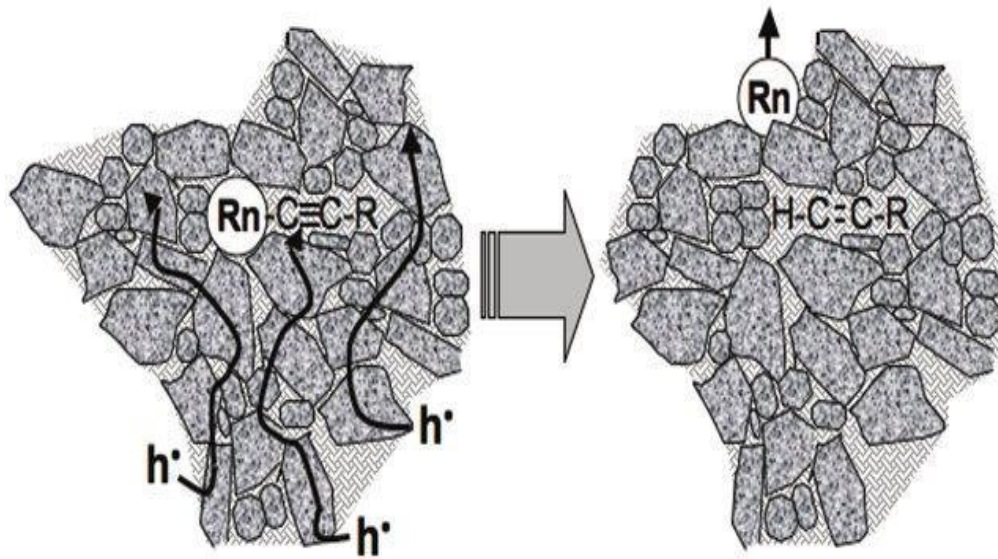


Figure 20 : Positive holes and emission of radon gas. [47]

It is perceived in some experiments that emission of p holes at stressed surface displays a strong peak and pulse like dynamics which can be result of the rapid discharge of ions into the air by the fracture surface [48].

To understand physics of this ions propagation we can consider the case of spherical capacitor where one plate is the surface of Earth and other plate is the lower edge of the Ionosphere. The potential difference of this heavenly capacitor is about 270 kV, providing effecting channel for airborne ions. These ionizing situations within the capacitor affects entire atmosphere and perturb the TEC [28], [49].

Another explanation also shows agreement among researcher that is from Solid State Physics we know that when the flood of positive charges is very prominent in rock it also changes the physical properties such as electrical conductivity of soil layer.

Carbon and Radioactive Radon are generally present in the composition of soil; this radon is usually trapped in the triple-bonded organic compounds these compound do this job very effectively [50]. Freund (2010) concluded that because of the stress induced and highly oxidizing positive holes, the triple bond is oxidized to double which make them incapable to retain radioactive radon any further. The allow the radon to immediately discharge from soil and reach surface where it can react with water, increase the surface temperature as it has short half-life. Such release of radon in the breeding zone of future EQs can be a very useful signal in EQ prediction

During Earthquake preparation the anomalous increase in radon and natural radioactivity show anomalies increase in positive holes on the surface[28], these escalating situations become strong enough, a strong electric field developed between lithosphere and ionosphere accelerates free electrons to trigger corona discharges. These accelerating ionize the air born gas molecules to produce negative air ions. The long duration abnormal values of SO_2 and SO_4 may be explained by these negative air ions.

5.2 Conclusion

In this paper, we have studies GPS-TEC fifteen days before and twenty five days after the main shock results shows that during normal geomagnetic conditions positive ionospheric anomalies also occur few days before EQ which might be due lithosphere-ionosphere coupling and negative anomalies occur after the EQ which is due to geomagnetic storm. Tropospheric delay experience by the GNSS receiver is also analyzed and EQ induced anomalies have been detected in GNSS Tropospheric wet delay within seismogenic zone shortly before the Earthquake. Furthermore, 2 hour daily average from 2 UT to 4 UT for several stations has also showed some clear anomalies. OLR and Geopotential Height supports these abnormal variations in tropospheric delay this conclude that might be the result of anomalous electric field induced by stressed-rocks in seismogenic zones and due to emission of radioactive gases like radon. The analysis is also supported by lower atmosphere changes in SO_2 and SO_4 . In future, more in depth analysis with multiple GNSS and Remote Sensing satellites are needed to clarify the phenomena is needed.